Fiscal Year:	FY 2017	**Task Last Updated:**	FY 07/14/2017
PI Name:	Masson, Patrick H. Ph.D.		
Project Title:	Using Brachypodium distachyon to Investigate Monocot Plant Adaptation to Spaceflight		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology (2) Developmental Biology		
Space Biology Special Category:	(1) Bioregenerative Life Support		
PI Email:	phmasson@wisc.edu	**Fax:**	FY 608-262-2976
PI Organization Type:	UNIVERSITY	**Phone:**	608-265-2312
Organization Name:	University of Wisconsin Madison		
PI Address 1:	Laboratory of Genetics (Room 3262)		
PI Address 2:	425G Henry Mall		
PI Web Page:	https://		
City:	Madison	**State:**	WI
Zip Code:	53706-1580	**Congressional District:**	2
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2014 Space Biology Flight NNH14ZTT001N
Start Date:	09/12/2014	**End Date:**	09/11/2018
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	5	**Monitoring Center:**	NASA KSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 9/11/2018 per NSSC information (Ed., 12/13/17) NOTE: End date changed to 9/11/2017 per NSSC information (Ed., 6/14/16)		
Key Personnel Changes/Previous PI:	No changes.		
COI Name (Institution):			
Grant/Contract No.:	NNX14AT23G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Most major cereal grain crops are monocots. Yet, most investigations of plant adaptation to the spaceflight environment have been carried out on the dicotyledonous model plant Arabidopsis thaliana. It remains unknown whether the conclusions from such studies can be extrapolated to monocotyledonous plants. To fill this knowledge gap, we propose to investigate the growth, development, and transcriptome profiles of Brachypodium distachyon seedlings under spaceflight conditions, and compare these adaptive responses to those recently documented for Arabidopsis seedlings. These investigations will lead to a better understanding of the molecular and developmental mechanisms that contribute to Brachypodium adaptation to spaceflight conditions. In the long term, these data may also lead to the development of better mitigation strategies aimed at improving monocot adaptability to spaceflight parameters.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Because most cultivated crops are monocotyledonous plants, there is tremendous interest in better understanding the molecular mechanisms that govern their growth behaviors and adaptation to stress. In addition to contributing to a better understanding of the molecular mechanisms that modulate plant adaptation to the microgravity environment, an important step toward engineering crops that are better suited to serve as components of the bioregenerated life-support systems during spaceflight, our work with Brachypodium distachyon should provide new insights into the molecular mechanisms that control root growth and development. Because root growth and its architecture have recently been recognized as key contributors to plant productivity and adaptation to environmental stress, our research is also likely to have long-term impact in agriculture, horticulture, and forestry. Furthermore, many of the behaviors under investigation involve auxin-related processes. This plant hormone is a key contributor to most aspects of plant growth, development, and response to the environment. Therefore, this project is likely to yield important new insights into some of the key mechanisms that control plant growth, development, and response to biotic and abiotic stresses.
Task Progress:	Plants are viewed as key components of bioregenerative life-support systems to sustain long-term space-exploration missions. Therefore, there is a need to better understand their morphological and molecular responses to the microgravity environment to effectively implement such systems. Because most cultivated crops on Earth involve monocots, there is interest in better understanding the molecular mechanisms that govern their responses to microgravity. Yet, to date most experiments aimed at investigating plant adaptation to spaceflight have involved Arabidopsis thaliana, a dicot. To improve our understanding of the mechanisms that govern monocot plant adaptation to spaceflight, we are preparing for experiments aimed at investigating the morphological and transcriptomic responses of Brachypodium distachyon seedlings to the microgravity environment of the International Space Station (ISS). The planned experiments will involve growing Brachypodium seedlings from three distinct accessions within the VEGGIE growth chamber on the ISS, and comparing their growth behaviors and transcriptomic profiles to those of ground-based controls (subjected to 1g). To be fully informative, the planned experiments will require comparing the morphology and transcriptomic profiles of Brachypodium seedlings exposed to either microgravity or 1g during their entire growth period, from germination to harvesting. This implies that the microgravity-exposed plants will have to derive from seeds that germinated after transfer into VEGGIE on ISS, under microgravity, or on Earth in the ground control. As reported last year, we tested two experimental approaches allowing to plate seeds on growth substrate on the ground, before flight, and transporting the planted material to ISS for germination and growth in VEGGIE. In the first protocol, Brachypodium seeds are plated on agar-containing medium within Petri dishes. The seeds are prevented from germinating prematurely by continuous blue-light treatment in the cold (4oC). Upon transfer into VEGGIE on ISS, germination can be induced by exposing the dishes to red light at room temperature for 24-36 h. Unfortunately, this system was deemed difficult to implement during spaceflight because it would require a significant power source to light up the plated seeds and keep the environmental temperature at 4oC during flight. Considering this limitation, an alternative protocol was developed under advice from Dr. Howard Levine, with technical assistance from the Kennedy Space Center Support Team. In this protocol, up to 24 surface-sterilized Brachypodium seeds are embedded in a block of dry Oasis Foam contained within a Magenta Jar. Because both foam and seeds are dry, the plant material remains dormant during vehicle takeoff, travel to ISS, and docking. After transfer into VEGGIE, astronauts will activate seed germination by injecting liquid medium into the foam and exposing the jars to red light for 24-36 hours. Subsequently, green and blue LED lights will be turned on, and the seedlings will be allowed to grow for a total of 5 days. The seedlings will then be photographed, harvested, transferred into Kennedy Fixation Tubes (KFTs), and fixed in RNALater for 24-36 h at room temperature. The fixed material will then be frozen at -70oC and stored until return to the ground. This year, we have demonstrated the effectiveness of this medium-injection system to trigger germination. We also designed a new medium injection device that fits within a modified Magenta Jar, and allows effective injection of medium into the foam without opening the jar. This system is designed to include a connecting spout that protrudes through a hole at the bottom of the Magenta jar. This spout allows easy snap-in connection to a hose attached to a growth medium-filled syringe. With this system it should be easier for the astronauts to connect a medium-filled syringe to the Magenta Jar spout and inject a defined amount of liquid into the foam, using the syringe. To allow more homogenous distribution of liquid within the foam, a layer of gauze is sawed on the surface of the foam, itself surrounded by a layer of Nitex membrane that prevents roots from penetrating the foam block. Seeds are embedded into the foam through slits cut within the Nitex and gauze surfaces. Seeds are stabilized within the slits by a drop of guar gum, which prevents them from dispersal by the vibrations typically encountered during vehicle takeoff. An adapter platform will also be designed to anchor up to 12 Magenta jars within VEGGIE (four biological repeats for each one of the three accessions: Bd21, Bd21-3, and Gaz-8). A Science Verification Test (SVT) was carried out at Kennedy Space Center on June 12-14, 2017. After seed planting into the dry foam and medium injection, the Magenta jars were transferred into a VEGGIE unit placed in a reach-in growth chamber programmed to reproduce the temperature, humidity, and CO¬2 profiles encountered on ISS. At the end of the growth period, Brachypodium seedlings were photographed, harvested, and fixed in RNA later within KFTs. They were frozen, and then returned to the Principal Investigator (PI) laboratory for molecular analysis. Frozen fixed seedlings were dissected, generating separate shoot and root samples that were processed for RNA extraction. These extracted RNAs are currently being analyzed for quality and ability to serve as effective substrates for RT-PCR analyses. In parallel, the pictures of seedlings grown in the Magenta Growth Chamber (MGC) are being analyzed to determine germination rate and organs growth. In addition to preparing for the APEX-06 spaceflight experiment, we have also continued our analysis of Brachypodium

	root gravitropism and growth behavior on agar surfaces. Last year, we reported on the development of time-lapse video imaging of Brachypodium root growth to evaluate the kinetics of root gravitropism and quantify the oscillatory growth behavior displayed by the roots of 42 distinct accessions developing on agar-based media. This analysis allowed us to identify several candidate loci that were associated with quantitative traits relevant to gravitropism and oscillatory root growth. This year, we have continued these analyses, developing a more robust imaging platform for automated analysis of seedling root growth behavior, and repeating our analysis on a larger number of seedlings for each accession. We have also acquired 60 additional Brachypodium accessions with fully sequenced genomes (kindly provided to us by Dr. John Vogel, University of California-Berkeley), which we are propagating under our own growth conditions for subsequent genome-wide association studies (GWAS) analysis. Furthermore, we initiated an analysis of primary root growth in the Bd21 accession, with the objective of mapping the meristem and elongation zones along the root tip for this accession, and also evaluating the relative contribution of each root-tip zone to the gravitropic response as well as the curvatures associated with oscillatory growth and autostraightening. A morphometric analysis of cell sizes along the root tip along with an analysis of the spatial profile of velocities using the Stripflow software (developed by Dr. Tobias Baskin, University of Massachusetts, Amherst, MA), are underway to evaluate the spatio-temporal distribution of curvatures along the root tip, hence identify key growth processes that are targeted by the gravity stimuli and/or the autostraightening response. Furthermore, transgenic lines expressing auxin-level and auxin-activity reporters, along with several reporters of auxin efflux facilitator localization (all kindly provided to us by Dr. Devin O'Connor, University of Cambridge, UK) are being used to evaluate the potential contribution of auxin redistribution and response to the various phases of root gravitropism and root oscillatory growth. Together, this combination of physiological, genetic, and cell biological investigations will provide new insights into the molecular mechanisms that modulate root gravitropism and oscillatory growth in monocots in general, and in Brachypodium distachyon in particular. In the longer term, these studies will yield novel information that may facilitate the development of novel breeding strategies or cultivation approaches to improve crop productivity on Earth and during spaceflight, considering the relevance of root growth behavior and root system architecture to plant productivity and survival under challenging conditions. The following publication is in production: Books/Book Chapters: Gibbs, N.M., Vaughn Rouhana, L., and Masson, P.H. "Quantitative Trait Loci for Root Growth Response to Cadaverine in Arabidopsis. " in "Plant Polyamines: Methods and Protocols." Ed. Alcazar, R., and Tiburcio, A. Springer, expected publication Dec-2017
Bibliography Type:	Description: (Last Updated: 07/03/2024)
Articles in Peer-reviewed Journals	Su S-H, Gibbs NM, Jancewicz A, Masson PH. "Molecular mechanisms of root gravitropism." Curr Biol. 2017 Sep 11;27(17):R964-R972. Review. https://doi.org/10.1016/j.cub.2017.07.015 ; PubMed PMID: 28898669 , Sep-2017